

Linu Anil Teresa

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EDUCATION

University of Maryland

Master of Science, Data Science

College Park, MD

Sept. 2025 – Expected May 2027

Saintgits College of Engineering

Bachelor of Technology, Computer Science and Engineering

Kerala, India

July 2019 - June 2023

EXPERIENCE

Backend Developer

Accenture Song

July 2023- July 2025

Kerala, India

- Built and maintained Spring Boot microservices that powered analytics dashboards for e-commerce clients serving 10K+ users daily, improving system response time by 35-40%.
- Automated API testing and data validation pipelines with Karate and JUnit, raising test coverage from 42% to more than 85% cutting QA turnaround by 60% and boosting release cadence.
- Integrated SonarQube quality gates and continuous integration pipelines on AWS/Azure, reducing production defects by 60%.
- Collaborated with cross-functional teams to analyze data performance and implemented optimizations at the microservice level that improved reliability and data synchronization.

PROJECTS

Multimodal AI and Multi-Agent Food Recommendation System

Nov.2025- Dec. 2025

- Developed a multimodal AI application converting unstructured food images into location-aware restaurant recommendations using chained LLM calls.
- Implemented a multi-agent decision system (Pro, Con, Judge agents) to evaluate trade-offs and generate preference-aware recommendations.
- Engineered an API-driven inference pipeline integrating Yelp Business APIs that normalize and rank restaurants based on ratings, reviews, availability, and real-time metadata.
- Reduced latency for AI responses by implementing asynchronous execution and robust fallback logic within FastAPI.

Predictive Analysis on Formula-One Overtaking rates

Aug. 2025 - Dec. 2025

- Formulated a quantitative metric (CODI - Circuit Overtaking Difficulty Index) to objectively quantify F1 circuit overtaking difficulty from the analysis of normalized historical race data.
- Engineered features from raw race logs and performed extensive data wrangling in Pandas to integrate and clean race results, circuit characteristics and FIA regulation changes.
- Implemented unsupervised learning (K-Means Clustering) to segment circuits into distinct difficulty tiers and validated the metric's stability using supervised classification models.
- Analyzed the impact of regulatory changes (e.g, DRS, weight adjustments, aerodynamic rules) on overtaking patterns, highlighting the project's applicability to policy evaluation and performance optimization.

INTERESTS AND SKILLS

Technical Skills: Python, Java, Springboot, Spring MVC, Unit Testing, Karate API testing

Data Science: Pandas, NumPy, Scikit-learn, Data-visualization, Exploratory Data Analysis, Feature Engineering

Cloud and DevOps: AWS, Azure, Terraform, Docker, SonarQube, Git, Jira, Bitbucket

Databases: SQL, MongoDB

Activities: TA (Teaching Assistant) at IIITH, Boot camp leader at Saintgits, participant in several National Hackathons

Interests: App Development, Data Science, Data Visualization, LLMs, Machine Learning, Predictive analysis